UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION
SEMESTER II - 2019/2020
PH 102 - General Physics II (3 credits)

Time allowed: ***Three** hours*

Answer **FIVE** questions selecting **TWO** from section-**A**,
TWO from section-**B** and **ONE** from section-**C**.
(Use of calculators is allowed)

All the symbols have their usual meanings unless specified otherwise.

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Mass of electron (m_e)	$= 9.11 \times 10^{-31}$ kg
Mass of proton (m_p)	$= 1.67 \times 10^{-27}$ kg
Permittivity of free space (ε_o)	$= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_o)	$= 4\pi \times 10^{-7}$ H m^{-1}
Planck's constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Stefan-Boltzmann constant (σ)	$= 5.67 \times 10^{-8}$ W m^{-2} K^{-4}
Avogadro number (N_A)	$= 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant (k)	$= 1.38 \times 10^{-23}$ J K^{-1}

Section-A

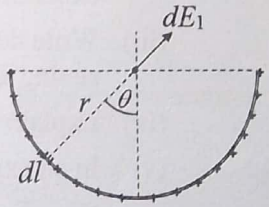
1. (a) An infinitesimal charge dq is at the origin in the free space. Write down an expression in vector form for the electric field intensity $d\vec{E}$ at a point given by position vector $\vec{r}$.

   (b) An insulating wire of length l is bent to form a semicircle as shown in the figure. If the wire has a uniformly distributed charge Q, obtain an expression for

   (i) electric field intensity at the centre of the semicircle and

   (ii) the potential at the centre.

   [**Hint:** *Take infinitesimal charge dq in length* $dl = r.d\theta$ *and consider the symmetry of charge distribution.*]



   (c) If the length of the semi-circular wire is 5.0 cm and the potential at the centre is 2.0 V find the total charge in the wire.

2. (a) A charge particle of mass m and charge $+q$ moves at speed v perpendicular to a uniform magnetic field of flux density $\boldsymbol{B}$. Obtain an expression for the radius of the path of the particle in terms of above parameters.

   (b) What is the period of the motion of the particle in part (a)?

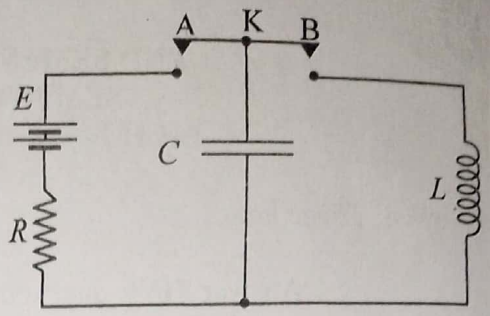
   (c) Draw a clearly labelled diagram of a cyclotron.

   (d) The maximum radius of a cyclotron is 80 cm and the magnetic flux density is 1.5 T. Find the cyclotron frequency necessary to accelerate deuterons of mass 3.3×10^{-27} kg and charge 1.6×10^{-19} C from this cyclotron.

   (e) What is the maximum kinetic energy (in eV) of the beam of deuterons that can be obtained from the cyclotron in part (d)?

CamScanner

3. A capacitor of capacitance C, a resistor of resistance R, inductor of inductance L, a two-way key K and a cell of e.m.f. E with negligible internal resistance are connected as shown in the figure. Initially, the capacitor is charged by pressing end A of the key K at time $t = 0$.



(a) For the charging of the capacitor,

(i) obtain an expression for the charge q in the capacitor after time t,

(ii) write down an expression for the time constant (τ) during charging.

(iii) Find the charge as a percentage of the maximum charge in the capacitor after 5τ time.

(b) Suppose $E = 6$ V, $R = 1.0$ kΩ and $C = 2.0$ μF.

(i) What is the time constant of charging?

(ii) Calculate the maximum energy that can be stored in the capacitor.

(c) When the capacitor is fully charged, end B of the key K is pressed. Without any proof, sketch the variation of voltage with time **and** current with time. Assume there is no resistance in the discharging circuit.

(d) If $C = 2.0$ μF and $L = 80$ mH find the frequency (in Hz) of the oscillator circuit.

Section-B

4. (a) (i) What is Compton Effect?

(ii) With the help of diagrams, explain the results of the Compton scattering experiment.

(iii) Write down the formula for the Compton shift identifying all the symbols and stating all the assumptions made.

(iv) Explain why there is no wavelength shift in some of the scattered photons.

(v) In an experiment, X-rays with wavelength 71 pm is incident on a calcite target.

(1) Find the wavelength of the X-rays scattered at an angle of 30°.

(2) What is the highest wavelength shift in this experiment?

(b) (i) State de Broglie hypothesis.

(ii) Find the de Broglie wavelength of following charged particles.

(1) An electron moving at a speed of 1.00×10^6 m s^{-1}.

(2) A proton with kinetic energy of 5 MeV.

(3) An electron with 50 GeV energy.

(iii) The neutrons produced in a reactor are known as thermal neutrons when their kinetic energy is reduced (by collisions) to the order of $\frac{3}{2}kT$.

(1) What is the kinetic energy of thermal neutrons if $T = 300$ K?

(2) What is the de Broglie wavelength?

(3) Describe an application of neutron diffraction.

CamScanner

5. (a) (i) What is Photoelectric effect?

(ii) Write down Einstein's photoelectric equation.

(iii) When light of wavelength 600 nm falls on a metal surface, photo-electrons with the maximum kinetic energy of 0.17 eV are emitted.

(1) Determine the work function of the metal.

(2) Find the cut-off frequency of the metal.

(3) What is the stopping potential when the above metal surface is illuminated with light of wavelength 400 nm?

(b) A star with a radius 8.5×10^{8} m has an emission spectrum with a peak at the wavelength of 685 nm.

(i) Find the energy of a photon corresponding to this wavelength.

(ii) What is the surface temperature of the star?

(iii) At what rate is the energy emitted from the star in the form of radiation? (Assume that the star is a black body)

$(C = 2.89 \times 10^{-3}\ mK)$.

6. (a) (i) Write down an expression for energy of an electron in the nth level of a hydrogen atom based on the Bohr model.

(ii) Calculate the change in energy corresponding to the following transitions of the hydrogen atom.

(1) From $n = 3$ to $n = 4$.

(2) From $n = 2$ to $n = 1$.

(3) From $n = 3$ to $n \rightarrow \infty$.

(b) (i) Show that the radioactive decay law can be given by $N = N_o e^{-\lambda t}$.

(ii) Define half-life $T_{1/2}$ and obtain an expression for $T_{1/2}$ in terms of λ.

(iii) The half-life of a radioactive isotope $^{198}_{79}Au$ is 2.70 days.

(1) What is the decay constant of $^{198}_{79}Au$?

(2) What is the probability that any $^{198}_{79}Au$ nucleus will decay in one second?

(3) Calculate the activity of 1.00 μg sample of $^{198}_{79}Au$.

(4) What is the activity of the above sample after one week?

CamScanner

Section-C

7. (a) Potential energy between two atoms separated by a distance r in a lattice can be written as $V(r) = -\frac{a}{r^m} + \frac{b}{r^n}$ where a and b are constants and m and n are positive integers.

(i) Identify the attractive and repulsive terms of the above expression and comment on their origin.

(ii) Prove that the separation between two atoms at minimum potential energy is given by $r_0 = \left(\frac{bn}{am}\right)^{\frac{1}{(n-m)}}$ for $n > m$.

(iii) Sketch the variation $V(r)$ versus r.

(b) Consider the rotation of a diatomic molecule around its centre of mass. The magnitude of molecule's angular momentum is given by $L = \sqrt{j(j+1)}\hbar$ where $j = 0, 1, 2$ .... is the rotational quantum number.

(i) Show that the allowed values of the rotational kinetic energy are given by,

$$E_{rot} = \frac{j(j+1)\hbar^2}{2I}$$

where I is the moment of inertia about the axis of rotation.

(ii) A gas consisting of identical diatomic molecules in which one molecule is in the $j = 0$ rotating state and the other molecule is the $j = 2$ rotating state. Assume that gas absorbs electromagnetic radiation over a wide range of frequencies. Due to this molecule in the $j = 0$ state makes a transition to $j = 1$ state while the molecule in the $j = 2$ state makes a transition to $j = 3$ state. What is the ratio of the frequencies of the photons that excited these molecules?

8. In general, molecules vibrate and rotate simultaneously. The total energy due to vibration and rotation of a molecule is given by,

$$E_{(v,j)} = \left(v + \frac{1}{2}\right) hf + \frac{\hbar^2}{2I} j(j+1)$$

where I is the moment of inertia, v and j are the vibrational and rotational quantum numbers respectively.

(a) Draw a diagram describing the energy levels of a molecule of two states with quantum number $v = 0$ and $v = 1$. Indicate sub energy levels for the quantum number $j = 0,\ 1,\ 2,\ 3$ and 4 in each energy level.

(b) Draw possible transition lines on the above energy level diagram.

(c) Consider the transition of a molecule from energy level $E_{(0,0)}$ to $E_{(1,1)}$ due to an absorption of photons in the IR region.

(i) What are the energy values corresponding to the levels $E_{(0,0)}$ and $E_{(1,1)}$?

(ii) What is the minimum photon energy required for the above transition?

****

CamScanner